

RESEARCH ARTICLE

Firm Performance on Artificial Intelligence Implementation

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ABSTRACT

In recent years, artificial intelligence (AI) has become a focal point in academic and business research. With breakthroughs in learning algorithms, AI applications in business operations are increasingly practical and impactful. AI offers tools for market analysis, decision-making support, and innovations in business models and processes, presenting a significant turning point for firms. Despite this, questions remain about whether AI implementation yields measurable business value or is merely a trend, challenging enterprises and managers. This study provides a significant contribution by empirically examining AI impact on firm-level performance through three key indicators: financial performance, productivity, and market value. Drawing on internal financial perspectives, this research reveals that while AI adoption enhances financial performance and market value, the advantages for AI first movers and better performers are not uniformly positive across all indicators. This nuanced analysis offers managers and stakeholders a deeper understanding of the tangible value of AI, guiding more informed implementation strategies.

JEL Classification: C58, G32, M15

1 | Introduction

In recent years, significant attention has been devoted to companies' investments in information technology (IT) (Carr 2003; Rettig 2007). With the development of IT, it has become possible to turn the capabilities of organizations and industries into competitive advantages. Managers and operators have long evaluated the value of IT and hoped that their investments in IT could generate authentic benefit; we call this idea the business value of IT (BVIT). The term BVIT is often used to refer to the impact of IT on organization performance, including productivity, profitability, cost reduction, competitive advantage, inventory reduction, and other performance indicators (Devaraj and Kohli 2003; Hitt and Brynjolfsson 1996; Kauffman and Kriebel 1988). Accordingly, BVIT helps managers and operators comprehend

whether investment in IT is worthwhile or not so that they can be responsible for their stakeholders and outside shareholders.

In order to find out the benefits of BVIT, many scholars and researchers have investigated BVIT-related issues. For example, Hitt, Wu, and Zhou (2002) studied the business impact of enterprise resource planning and its productivity measurement. Bharadwaj (2000) employed a resource-based perspective to study IT capability and firm performance. Hitt and Brynjolfsson (1996) used three measures to investigate the relationship between IT stock and profitability ratios. Dos Santos, Peffers, and Mauer (1993) focused on announcements of IT investments and innovative IT investments. Overall, the eventual goal of these studies has been to understand whether IT can help firms to increase productivity. The research methods usually

use a well-known production function, Cobb–Douglas, to estimate the profitability of enterprises through some indicators, and calculate the market value by Tobin's q as well (Brainard and Tobin 1968).

Nowadays, artificial intelligence (AI) has become the most popular topic in several industries. AI is more than just a buzzword; its adoption in enterprises represents a growing trend, a well-established strategy, and an innovative approach to business management. John McCarthy in 1956 first proposed the concept of AI and described that an attempt would be made to find how to lead machines to use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. Recently, the best-known and successful cases of weak AI are AlphaGo and Watson Health (weak AI, in contrast to strong AI, cannot attempt to perform the full range of human cognitive abilities). These AI systems rely on the machine learning algorithm of called “Deep Learning.” Therefore, by implementing new AI technologies, firms could improve business productivity, provide continuous innovation for products and services, and expect high growth income in the future. Makridakis (2017) predicted that AI would dominate human life 20 years from now and generate an unavoidable shift in education, management, and employment.

While substantial research has examined BVIT, particularly with respect to traditional IT investments, the specific impacts of AI implementation on firm performance remain less explored. Previous studies have generally assessed BVIT in terms of productivity, profitability, cost reduction, and competitive advantage, often focusing on IT investments that support established operational processes. The unique characteristics of AI, however, including its ability to enhance decision-making through machine learning and to foster innovation in business models suggest that AI impact on firms may differ from that of conventional IT.

Despite the growing enthusiasm for AI adoption, gaps exist in our understanding of its real business value. For instance, while AI is anticipated to boost productivity and financial performance, empirical evidence of these outcomes remains sparse, particularly at the firm level. In addition, previous studies on first-mover advantages in IT adoption have shown mixed results, with some research suggesting that early adopters gain competitive benefits, while others indicate no significant advantage. Finally, the impact of AI on market value, especially over the long term, has been underexplored, leaving managers uncertain about return on investment in AI and its broader financial implications.

To address these gaps, this study presents three indicator (dummy) variables, including AI adopters, first movers, and better (AI) performers. Through quantitative analysis of our collected data and the secondary data, we thoroughly investigate the impact of AI based on these indicators. IT-type firms have their own AI abilities and rely heavily on IT. This could lead to inaccurate measurement of performance. Therefore, this study does not include IT-based companies in our samples (Huang, Lee, and Chou 2020). The arguments presented here are based on the works of Hitt, Wu, and Zhou (2002), Lieberman and Montgomery (1988), and Huang, Lee, and Chou (2020). The

results reveal that AI adoption has great impacts on financial performance and market value for AI adopters. Moreover, AI first movers and better performers are positively impacted according to the three indicators.

2 | Related Works

In this section, we discuss two areas of research literature. One is the business value of information technology (BVIT) in Section 2.1 and the other is AI in Section 2.2.

2.1 | Business Value of Information Technology

As the first step, we should figure out the definition of the business value of information technology (BVIT). Devaraj and Kohli (2003) considered empirical BVIT by focusing on “studies that are at the organizational level and include one or more IT-related independent variables and one or more dependent variables related to IT's organizational impact.” There are other related definitions. Schryen (2013) defined the business value of information systems (IS) as “the impact of investments, in particular, IS assets on the multidimensional performance and capabilities of economic entities at various levels, complemented by the ultimate meaning of performance in the economic environment.” Kohli and Grover (2008) pointed out that IT value manifests itself in many ways, including productivity gains, process improvements, profitability enhancement, increased consumer surplus, and improvements in supply chains or innovation at the inter-organizational level. Melville, Kraemer, and Gurbaxani (2004) defined BVIT as “the organizational performance impacts of IT at both the intermediate process level and the organization-wide level and comprising both efficiency impacts and competitive impacts.” Since the definition of Devaraj and Kohli (2003) is simple and explains causal idea, we adopt their definition in this study. Managers are long to evaluate the business or economic value of IT; therefore, the research result of BVIT can be attributed to response to the productivity paradox (Brynjolfsson 1993).

In general, managers expect that the investment in IT should result in payoffs for their companies. However, when it comes to the relationship between productivity and IT, past research has not found a positive relationship between productivity (or profitability) and expenditures. In spite of these uncertain results, Erik Brynjolfsson (2017) found that firms with the highest IT capabilities have significantly higher market values compared to firms with the lowest IT capabilities. In addition, some previous studies demonstrated that IT investment has assuredly advanced organizational performance (Devaraj and Kohli 2003; Hitt and Brynjolfsson 1996; Mukhopadhyay, Kekre, and Kalathur 1995). As explained above, it is clear that most of these studies suggest that the introduction of IT can bring positive benefits.

Regarding the subject of BVIT, two survey streams have investigated two BVIT levels, industrial (or economic) and firm levels. The former is a widely observed level, whereas the latter is a seldom observed level. The reason to study the former is because Robert Solow, the Nobel Prize laureate and economist, stated in the New York Times Book Review (July 12, 1987) that we see

the computer age everywhere except in productivity statistics. By the late 1980s, the conventional wisdom was that computers could not contribute significantly to productivity. Therefore, scholars opened the field to address this issue.

The second facet is observed at a narrow firm level, where “there is no correlation whatsoever between expenditures for IT and any known measure of profitability” (Strassmann 1997). The early studies established a result that either no relationship or a slightly negative relationship exists between firm-level spending on IT and firm performance (Dehning and Richardson 2002). As a result, IT, accounting, or financial researchers have begun to turn research questions from productivity paradoxes (returning from IT) to economic measures (why or when there is a return from IT). By the end of the 1990s, some studies found that IT investments have a large return (Dehning and Richardson 2002).

Overall, the two levels are always able to obtain data from the observed or secondary datasets (Masli et al. 2011). The measurement of dependent and independent variables requires not only general accounting data from firms but also IT-related statistics from firms, such as IT inventory, computer capital, or IT labor costs. In this study, we adopt the firm level because most researchers have dealt with firm-level processing for various IT-related systems or technologies. Furthermore, it is easier to collect data for the firm level than for the industry level.

For the most part, researchers often use two methods to survey the impact of IT investments (business performance). The first is to understand the short-term market reaction of investors by event research methods. The effective market theory provides the theoretical cornerstone for this basic event research method (Malkiel and Fama 1970). Using this method, the manager can understand whether a serious accident affects the stock market. Numerous studies take advantage of this approach in various fields, such as economics and finance (MacKinlay 1997), management (McWilliams and Siegel 1997), and IT operational risk (Goldstein, Chernobai, and Benaroch 2011).

Furthermore, the second method is to measure accounting performance by accounting (financial)-based methods, such as gross margin, inventory turnover, cash flow, sales growth, and market share. This approach is also known as a backward-looking idea because it reflects the company's past performance. The measurement time and when the evaluation of IT investments occurred must be determined by the researcher (Masli et al. 2011). As mentioned earlier, many researchers have studied the firm level and investigated the relationship between a variety of IT investments and their BVITs.

As mentioned above, most previous research approaches can be divided into two methods, an event research method and an accounting-based method. Event research methods are used to objectively investigate the external validity of IT investments; however, accounting-based methods are used to objectively investigate the internal efficiency of IT investments. In this study, we focused on investigating the internal efficiency of IT investments. For the issue of the productivity and profitability for IT, the investigations of Dehning and Richardson (2002), Masli et al. (2011), Devaraj and Kohli (2003), and Tang, Huang, and Wang (2018) are representative and are worthy of reference.

2.2 | Artificial Intelligence (AI)

AI, the most discussed topic for companies nowadays, is one of indicators for companies' future development. The term “Artificial Intelligence” was proposed by John McCarthy (Dartmouth College) in 1956. AI was defined as an attempt made to find how to lead machines to use language, form abstractions and concepts, handle various kinds of problems now reserved for human beings, and improve themselves. AI has appeared in films and novels and has become a frequently discussed subject. In fact, AI services or products have been highly developed and integrated into our daily lives, even though it has been portrayed as part of a seemingly imagined future and fictional story.

As stated above, the current best-known product of AI implementation is AlphaGo (Google), which defeated Korea's top Go player Lee Sedol in 2016 and China's top Go player Ke Jie in 2017. Another famous AI product is Watson Health (IBM), which can be used to support diagnosing, treating, and predicting outcomes in many medical situations. In fact, most applications of AI are not examples of strong AI, but rather weak AI. The latter can be pinpointed as machine learning, of which the idea is to train supervised or unsupervised algorithms to do one special thing well, but not many general things. As a result, through the machine learning procedure of Watson Health, such as when studying numerous texts from research papers, electronic health record data and so on, Watson Health is able to provide a suitable medical treatment. For example, Watson Health can deal with numerous MRI images of the brain and circle any marks that look like a brain bleed for the doctor to examine (Captain 2017).

How can AI do it? These successes rely on an algorithm of machine learning called “Deep Learning.” In brief, this AI training process deals with a set of elements X, corresponding to a set of outputs Y. For example, the input element X may be photographs of a variety of different animals, and then the correct output Y may be the names of these animals, such as dogs, cats, and horses. Next, we use thousands, or even millions of training data to make the system learn. Finally, the correct rate of the successful AI system will increase with the amount of training data. A successful AI system can help users to quickly and correctly complete work. Further, AI can predict the next step to improve overall efficiency. For instance, JPMorgan Chase implements a new AI system that can complete the examination of commercial loan contracts in a matter of seconds, while it previously took lending officers 360,000 h to complete the job.

What is the revolutionary impact of implementing AI for enterprises? We can discuss two impacts. One is the impact on technology, and the other is that on decision-making. For the former, firms depend on the speed at which AI technologies can succeed in automating non-repetitive mental tasks currently performed by humans and replacing them in the process. AI programs can develop new computer programs by themselves to expedite the process of technological change. There is no doubt that AI technologies coupled with the Internet will affect the business models of firms and how they are managed, while also influencing employment patterns (Makridakis 2017). In addition, Wang et al. (2023) explored how AI and big data technologies are utilized in managing enterprise information security and

conducting risk assessments, providing insights into the role of AI in enhancing organizational security frameworks. However, the review examined potential negative implications of AI in electronic markets, offering a balanced perspective on AI adoption by highlighting associated risks and challenges (Xing et al. 2023).

For the latter, one way to materialize the synergistic relationship between AI and human beings is to combine the speed of AI in collecting and analyzing information with humans' superior intuitive judgment and insight. AI can furnish some utility computer programs that lead decision makers to overcome equivocal situations and address relevant conflicting needs. For example, AI systems that conduct sentiment analyses of internal and external channels of social media tend to provide a more precise reflection of possible reactions to organizational decisions (Jarrahi 2018). Such AI systems can improve the overall value of the business. For instance, Facebook's market cap per employee of \$22.1 million is more than nine times that of Johnson and Johnson (J&J), at \$2.43 million. In addition, even Facebook revenues per employee of \$1.15 million are more than ten times higher than those of J&J, which demonstrates the elevated productivity of Facebook's workforce. We are, therefore, able to understand why AI is one of imperative information technologies for numerous managers who are going to implement, implementing, or have implemented AI into their firms. To sum up, there are three reasons that firms are willing to implement AI technologies; they are (1) improving business productivity, (2) providing continuous innovation for products and services for consumers, and (3) expectations of high growth income of firms in the future. Accordingly, the three reasons may cause the stock prices of firms implementing AI to rise, explaining why managers are currently paying much attention to AI technologies (Makridakis 2017).

In this study, we assess AI financial performance, productivity, and market value through a financial/accounting-based approach. We use another perspective to systematically examine the internal validity of AI and understand the commercial value of AI.

3 | Research Hypotheses

As mentioned in the previous section (Section 2), it is clear that investing in IT has proven to be superior in BVIT. Similarly, since AI is also an IT investment, including algorithms, big data, and the Internet, we try to confirm that the investment in AI will lead to more benefits through various indicators. Therefore, the assumptions we propose are as follows:

Hypothesis 1a. *AI implementation in a firm will provide greater efficiency as measured by financial performance measurements.*

Hypothesis 1b. *AI implementation in a firm will provide greater efficiency as measured by productivity.*

Hypothesis 1c. *AI implementation in a firm will provide greater efficiency as measured by stock market value.*

The first firms to develop new products and services can usually occupy most of the previous market and gain a series of advantages, which is determined by the first-mover advantage in strategic management, with which pioneering companies can obtain positive economic profits (Lieberman and Montgomery 1988). According to the above discussion, the key factor of the first mover advantage is the "order of entering the market." On average, the first mover strategy brings a long-term competitive advantage to a company, since it can gain a large share of the product and service market and a higher profit than late competitors. According to Lieberman and Montgomery (1990), in order to obtain the first-mover advantage, there are three main methods: (1) buyer conversion costs, (2) technology leadership, and (3) asset preemption. Therefore, through technological growth, such as (1) innovative products or services and (2) new processes or process reengineering to consolidate competitive advantage, companies can enter a new market. Many companies implement new IT technologies to design new business models and improve business performance. As stated above, we believe that the implementation of AI technology may also bring the first-mover advantage to enterprises. Therefore, we have the following assumptions:

Hypothesis 2a. *Firms adopting the first-mover advantage of AI implementation will present better efficiency according to financial performance measurements.*

Hypothesis 2b. *Firms adopting the first-mover advantage of AI implementation will present better efficiency according to productivity measurements.*

Hypothesis 2c. *Firms adopting the first-mover advantage of AI implementation will present a better efficiency according to stock market value.*

A survey reporting from SAS of Accenture Applied Intelligence, Intel, and Forbes Insight studied how firms are benefitting from AI.¹ SAS indicated that firms which implement the AI-related technologies can save business-related cost expenditures, optimize business processes and efficiency, and finally improve profitability.

In this report, SAS revealed that AI has helped various companies to digitize their business process and find some amazing results. Three such results showing that these companies present actual efficiency after implementing AI technology include the accuracy of forecasting and decision-making increased by 60%, developing and liaison with customers efficiency increased by 52%, and productivity increased by 48% (Insights 2018). As mentioned above, a better (AI) performer means that companies in certain industries can introduce AI-related technologies, and the performance of the company should be better compared to other companies. Therefore, we set up the following hypotheses:

Hypothesis 3a. *Firms considered better performers will present better efficiency according to financial performance measurements.*

Hypothesis 3b. *Firms considered better performers will present better efficiency according to productivity measurements.*

Hypothesis 3c. *Firms considered better performers will present better efficiency according to stock market value.*

4 | Research Methodology

Three major research streams are included in this section: data collection and selection in Section 4.1, descriptive statistics in Section 4.2, and research methods in Section 4.3.

4.1 | Data Collection and Selection

The notion of AI was first proffered at the Dartmouth Society in 1956. However, the development of AI technologies has gained more attention after deep learning was proposed in 2010. Therefore, we try our best to collect all possible news, including companies who invested or implemented AI between 2010 and 2017. The data collection process is described as follows. The data we collected in this study are from 2010 to 2017; however, the baseline for performance analysis is only available in 2017. The baseline indicates that we only use the collected performance data for these companies in the 2017 fiscal year.

In this study, two programs were used to collect data. We confirmed the list of sample companies, collecting data through Compustat and then selected some ratios to evaluate their performance. First, the sample framework for this study is based on the S&P 500 Index list. The S&P500 Index is the US stock market index based on the market capitalization of 500 large companies, listed on the New York Stock Exchange (NYSE), Nasdaq Stock Exchange (NASDAQ), or Cboe BZX Exchange. This is one of the most common stock indices and many researches consider it to be the most representative of the US stock market. Based on this list, we first excluded companies whose industry category belongs to information technology, and then used Google to browse relevant news through a keyword, artificial intelligence, to see if

S&P 500 companies announced AI-related news. In addition, we used the Google search engine to check other keywords relating to AI, which could be employed to find AI related information. We also used them to gather all the stories about AI in the news. Our only concern was the huge impact of IT-type companies on IT; we aimed to measure the true influence on AI without this impact. Therefore, in order to ensure that companies with news mentioning AI were definitely implementing AI, we carefully read the news content. Finally, we collected 249 news stories, but we filtered out 152 news stories because they were about IT-related companies. The remainder (97 news) were included in our experiment.

After confirming the companies that had implemented AI, we executed the second procedure. The financial data of these AI companies were collected through the Compustat database. Compustat is a database of financial, statistical, and market information for effective and ineffective global companies, and includes various financial ratios. We used financial data to assess the companies' performance. The ratios (or indicators) we chose are listed in Table 1.

4.2 | Descriptive Statistics

To test the Hypotheses 1a, 1b, and 1c, we searched for companies on the S&P500 list that did not announce AI news; we used them

TABLE 2 | Sample statistic.

AI adopter	50	Behavior	First mover	23
			Second mover	27
AI non-adopter	250	Performance	Better	23
			Non-better	27
Total	300			

TABLE 1 | Financial performance measurement.

Ratio	Definition	Interpretation of high ratio
Profit margin	Pretax income/sales	Higher profit generated by sales
Labor productivity	Sales/number of employees	Higher productivity per employee
Asset turnover	Sales/assets	The speed of total assets turnover is faster, the sales ability is stronger, and the efficiency of asset utilization is higher.
Return on assets (ROA)	Pretax income/assets	The efficient use of corporate assets
Return on equity (ROE)	Pretax income/equity	Higher return on investment of corporate shareholders
Inventory turnover	COGS/inventory	High capital utilization and positive liquidity
Labor productivity	Sales/number of employees	Higher productivity per employee
Debit to equity	Debit/equity	Higher the risk to the company
Account receivable turnover	Sales/account receivable	The company mainly focuses on spot sales, or it is quite efficient in collecting payment.
Tobin's q	Market value/book value	Tobin's q > 1, which means that the market is rewarding the firm, and there is increased willingness of entrepreneurs to invest their capital or increase capital to set up factories.

for the purpose of comparison between two sample groups. We collected a total of 300 samples without IT companies; 50 of the companies announced AI news, and 250 did not (see Table 2). Furthermore, we only analyzed the performance data in 2017, so those samples which did not have relevant AI news from 2010 to 2017 were deleted.

For testing Hypotheses 2a, 2b, and 2c, we divided all of the samples (adopting AI) into two groups. There were 50 companies that announced AI related news from 2010 to 2017, and we found that more than half of the related announcements were unveiled in 2017. Hence, we consider the fiscal year of 2017 to be the benchmark. Corporations announcing the AI news before 2017 were defined as first movers; on the contrary, the others were treated as second movers. Therefore, the number of first movers is 23 and that of second movers is 27.

For validating Hypotheses 3a, 3b, and 3c, the full samples were divided into two groups, better AI performers and non-better AI performers. According to the Standard Industrial Classification Code (SIC), we divided the S&P 500 companies into six industries (see Table 3). The first group included retail (SIC = 50–59) and financial and insurance (SIC = 60–64), and the second group included manufacturers (SIC = 10–33, 45, 47, 49, 89), conglomerates (SIC = 48, 65, 99), and healthcare (SIC = 80).

In addition, we consulted a report from American Society for Quality (ASQ) to divide the companies into two groups (Post.com 2017). According to the report, retailing and financial services perform better than manufacturing and supply chain management. Therefore, the number of better performers is 23 and that of non-better performers is 27. Moreover, the ASQ survey reported that only 12% of all company cases had deployed AI commercially (Post.com 2017). Thus, the data we collected showed that 16% of companies had deployed AI commercially. In summary, the reported data are similar to the data we collected, and they also reveal that there are currently not many companies developing commercial AI.

4.3 | Research Methods

Ordinary least squares (OLS) regression method was used to analyze the impact of AI on performance. We utilize three fundamental descriptions, financial performance analysis,

productivity function, and stock market valuation (Tobin's q), for the analysis of the performance impact of AI adoption. On the basis of the study of Hitt, Wu, and Zhou (2002), we extended the method of financial performance analysis. However, to correspond with our hypothesis purpose, we have changed some research variables. What follows is a formula, which is the original form of the financial metrics with the logarithm function.

$$\text{Log}(\text{ratio}) = \text{Log} \frac{(\text{ratio numerator})}{\text{ratio denominator}} \quad (1)$$

To represent the general form of the estimated regression, we used the equation of $\text{Log}(\text{ratio numerator}) = \text{Log}(\text{ratio numerator}) - \text{Log}(\text{ratio denominator})$ as described above to form the following:

$$\begin{aligned} \text{Log}(\text{ratio numerator}) = & \text{intercept} + \text{Log}(\text{ratio denominator}) \\ & + \text{dummy variables} + \text{industry variables} + \varepsilon \end{aligned} \quad (2)$$

In this study, we consider the ratio numerator on the left as the dependent variable and the right side of the equation as an independent variable, including the ratio denominator, AI adoption (virtual variable), industry variables, and error terms. The composition of the formula allows us to provide greater flexibility in the relationship between the numerator and the denominator while maintaining an interpretation of the raw ratio performance. Therefore, on the premise of controlling industrial variables, we use the various financial performance indicators in Table 1 to observe different perspectives of AI performance.

As discussed by Hitt, Wu, and Zhou (2002), the diversified perspective of BVIT can be captured by a variety of financial indicators, which is an advantage of this analysis method. Despite this, the model lacks a strong theoretical foundation. Therefore, it should be interpreted as the result of correlation rather than the estimation of the econometric model. To reduce the uncertainty, there are two methods to measure the performance of AI adopters, productivity function and stock market valuation (Tobin's q) (see Table 1). Again, these two indicators can be used to track the investigation (Hitt, Wu, and Zhou 2002). Before applying these indicators to test the performance of AI adoption, we introduce the three methods, including financial performance analysis, productivity function, and stock market valuation (Tobin's q).

4.3.1 | Financial Performance Analysis

Financial performance analysis is a subjective indicator to measure assets and revenue to reveal a company's performance. It can be used to predict short-term and long-term business operations and growth; accordingly, it is also considered as a financial performance for firms. Therefore, regardless of corporate managers, operators and investors are concerned about the status of financial indicators. For business managers and operators, financial performance analysis can provide valuable information to make decisions. For investors, these indicators can be considered as an assessment of risk. In terms of financial performance, there are various indicators

TABLE 3 | Industry matching by SIC code.

Industry	N	Percentage	SIC code
Manufacture	104	35%	15–33, 45, 47, 49, 89
Financial and insurance	75	25%	60–64
Energy and materials	58	19%	10–14
Retail	43	14%	50–59
Conglomerates	15	5%	48, 65, 99
Healthcare	5	2%	80
Total	300		

for measuring financial performance; for example, we use the return on equity (ROE) indicator to check the profitability of a company's use of its own capital. In addition to ROE, there are many indicators that can be used for financial performance, as shown in Table 1. From Equation (2), the financial indicators in Table 1 are tested by three estimated regressions.

1. Hypotheses 1a, 2a, and 3a:

$$\begin{aligned} \text{Log (ratio numerator)} &= \text{intercept} + \text{Log (ratio denominator)} \\ &+ \text{AI adoption variables} + \text{industry variables} + \epsilon. \end{aligned} \quad (3-1)$$

$$\begin{aligned} \text{Log (ratio numerator)} &= \text{intercept} + \text{Log (ratio denominator)} \\ &+ \text{first mover variables} + \text{industry variables} + \epsilon. \end{aligned} \quad (3-2)$$

$$\begin{aligned} \text{Log (ratio numerator)} &= \text{intercept} + \text{Log (ratio denominator)} \\ &+ \text{better performer variables} + \text{industry variables} + \epsilon. \end{aligned} \quad (3-3)$$

In this subsection, we have explained how financial indicators can be used to calculate financial performance. We continue to discuss how to estimate productivity functions by using productivity indicators.

4.3.2 | Productivity Function

First, the Cobb–Douglas productivity regression is based on the economic concept of the production function. In economics, the production function explores the physical output and physical input or production factor of the production process. Because the advantages of regression are simplicity and empirical robustness, it is the most common and well-known approach to measuring production functions, which is used to calculate performance differences (Hitt, Wu, and Zhou 2002).

As mentioned earlier, we can understand that the key concept of the Cobb–Douglas function is to study the relationship between input and output during production. Therefore, we tend to use the Cobb–Douglas function to explore the impact of enterprise implementation of AI.

In this study, we refer to the output agent of the production relationship as sales and value added (VA) (VA = sales minus material), and the input substitution is fixed consumption, that is, capital (K) and labor (L), while controlling for time and industry. The log–log form is assumed because its coefficient can be interpreted as a percentage difference in productivity. Based on three different Cobb–Douglas regressions, three hypothetical groups are examined as follows:

2. Hypotheses 1b, 2b, and 3b:

$$\begin{aligned} \text{Log (VA)} &= \text{intercept} + a_1 \log K + a_2 \log NE + a_3 \log EE + \\ &\text{AI adoption variables} + \text{industry variables} + \epsilon. \end{aligned} \quad (4-1)$$

$$\begin{aligned} \text{Log (VA)} &= \text{intercept} + a_1 \log K + a_2 \log NE + a_3 \log EE + \\ &\text{first mover variables} + \text{industry variables} + \epsilon. \end{aligned} \quad (4-2)$$

$$\begin{aligned} \text{Log (VA)} &= \text{intercept} + a_1 \log K + a_2 \log NE + a_3 \log EE + \\ &\text{better performer variables} + \text{industry variables} + \epsilon. \end{aligned} \quad (4-3)$$

TABLE 4 | Introductions of variables.

	Variable	Code	Source
Input	Capital	K	Raw data from Compustat
	Numbers of employees	NE	Raw data from Compustat
	Employee expense	EE	Raw data from Compustat
Output	Value added	VA	Sales minus cost of goods sold

As described in Table 4, the data for the above formulas are collected from the Compustat database. Although the financial analysis method of the previous section can provide an analysis of the impact of enterprise AI implementation on the business from various aspects, adding productivity regression provides a more rigorous underlying approach to verify the impact of implementing AI on productivity. However, short-term measurements do not confirm the long-term effects of AI import. To overcome this weakness, we tested the performance of AI through stock market valuations. In the next section, we introduce related research methods.

4.3.3 | Stock Market Valuation

We assess the value of AI implementation by looking at the stock market to compensate for the shortcomings of the first two methods. It is important to note that we used the Tobin's q method. Tobin's q ratio was originally introduced by James Tobin in 1969 as a predictor of a company's future investment. Through the forward-looking Tobin's q approach, we predict the future development potential of companies' AI investment. It is widely regarded as indicative of future market prospects and reflects the contribution of IT to intangible value (Bharadwaj, Bharadwaj, and Konsynski 1999). Using Tobin's q, the three hypothesis groups are tested as follows:

3. Hypotheses 1c, 2c, and 3c:

$$\begin{aligned} \text{Log (market value)} &= \text{intercept} + a_1 \text{Log (book value)} \\ &+ \text{AI adoption variables} + \text{industry variables} + \epsilon. \end{aligned} \quad (5-1)$$

$$\begin{aligned} \text{Log (market value)} &= \text{intercept} + a_1 \text{Log (book value)} \\ &+ \text{first mover variables} + \text{industry variables} + \epsilon. \end{aligned} \quad (5-2)$$

$$\begin{aligned} \text{Log (market value)} &= \text{intercept} + a_1 \text{Log (book value)} \\ &+ \text{better performer variables} + \text{industry variables} + \epsilon. \end{aligned} \quad (5-3)$$

5 | Data Analysis and Results

5.1 | Comparison Between AI Adopters and Non-AI Adopters

Table 5 shows which indicators are significant, based on the financial performance regression formula (3–1). Different performance indicators were used to examine AI implementation enterprise (adopter = 1) and the indicator variables of industrial control variables.

TABLE 5 | Performance ratio regressions (AI adopters and non-adopters).

Interpretation	Labor productivity	Return on assets (ROA)	Inventory turnover	Return on equity (ROE)	Profit margin	Assets utilization	Collection efficiency	Leverage
Dependent variable	<i>In(Sales)</i>	<i>In(Pretax Income)</i>	<i>In(Cost of Goods Sold)</i>	<i>In(Pretax Income)</i>	<i>In(Pretax Income)</i>	<i>In(Sales)</i>	<i>In(Sales)</i>	<i>In(Debt)</i>
<i>Implementation</i>	0.118* (0.05)	0.027 (0.054)	0.064 (0.079)	0.08 (0.054)	0.081* (0.049)	0.059 (0.062)	0.143** (0.068)	0.136** (0.068)
<i>In(Employees)</i>	0.74*** (0.031)							
<i>In(Assets)</i>		0.641*** (0.035)				0.641*** (0.039)		
<i>In(Inventory)</i>			0.0678*** (0.037)					
<i>In(Equity)</i>				0.688*** (0.04)				0.695*** (0.049)
<i>In(Sales)</i>					0.072*** (0.039)			
<i>In(Account Receivable)</i>							0.534*** (0.031)	
Control variable				Industry				
<i>R</i> ²	0.594	0.499	0.599	0.511	0.556	0.404	0.266	0.559
<i>N</i>	298	275	275	266	275	300	296	285

**p* < 0.05.
***p* < 0.01.
****p* > 0.001.

TABLE 6 | Productivity and market value regressions (AI adopters and non-AI adopters).

Dependent variable	<i>In</i> (Gross Profit)	<i>In</i> (Sale)	<i>In</i> (Market value)
Interpretation	Value added	Output	Tobin's q
<i>Implementation (1 = AI Adopters)</i>	0.011 (0.066)	0.015 (0.042)	0.133** (0.052)
<i>In</i> (Ordinary Capital)	0.545*** (0.067)	0.303*** (0.043)	
<i>In</i> (Employees)	−0.008 (0.114)	0.209* (0.076)	
<i>In</i> (Labor Expense)	0.482*** (0.135)	0.22* (0.088)	
<i>In</i> (Cost of Goods Sold)		0.417*** (0.029)	
<i>In</i> (Total Assets)			0.626*** (0.032)
Control variable	Industry		
R^2	0.871	0.936	0.463
Observation	59	59	299

* $p < 0.05$.** $p < 0.01$.*** $p < 0.001$.

Overall, we found that when controlling for industry, AI adopters showed greater performances in terms of labor productivity, profit margin, collection efficiency, and leverage, amounting to 0.118, 0.081, 0.143, and 0.136, respectively. Even though the return on assets, inventory turnover, return on equity, and assets utilization were not significant, the positive effect (0.027, 0.064, 0.08, and 0.059) on the performance of AI still partially supports our proposed hypothesis. The results of historical internal financial data analysis show that adopting AI technology not only improves companies' operational capabilities and profitability, but also helps to improve the flexibility of the operating budget and create profits.

Table 6 shows the performance results of productivity and market value. Formula (4–1) is used to evaluate short-term productivity results. The results of the production function show that the value added is 1.1% and the output is 1.5%. Although AI had a positive but statistically insignificant impact on company productivity in the short term, our analysis did not uncover significant findings through statistical verification. Several other factors that may affect gross profit, including industry competition, cost structure, competitive products, expertise, innovation, and customers. Due to the timing of data collection, we were unable to prove the long-term performance of AI. Therefore, we used Tobin's q to capture the intangible value of companies' introduction of AI and the long-term returns of the stock market. The results indicate that the value of companies implementing AI was approximately 13.3%, surpassing that of non-AI companies.

Undoubtedly, AI can bring advantages and has a powerful influence on business management strategy and market value. As mentioned earlier, it is clear that the two proposed hypotheses are supported, Hypotheses 1a and 1c, indicating that companies with AI have better financial performance and market value. Although productivity measurements are not significant, they still have a positive impact on value added and output. Therefore, we still claim that Hypothesis 1b can be partially supported.

5.2 | Comparison Between First Movers and Second Movers

As mentioned earlier, first movers can achieve first-mover advantage after they publish new products or services. As a consequence, the purpose of the second hypothesis is to verify whether the order of entering the market affects the performance of the firm, which could become a competitive advantage. We used financial indicators to calculate the performance results of the first mover. The results of the regression analysis, according to the formula (3–2), can be observed in Table 7. The indicator items, including labor productivity, asset utilization, and collection efficiency, are the effects (−0.31, −0.061, −0.053). They had a negative relationship with the independent (virtual) variables. In short, the pioneer companies implementing AI showed no significant positive impact on financial performance, and almost half of the financial indicators indicate a negative impact on performance for first movers.

According to the results, the negative asset utilization (−0.061) result showed that the relevant resources were not effectively utilized, the related cost increased and the operating performance of the first mover decreased, resulting in a decrease in overall corporate performance, with a negative impact of about 6.1%. However, while the indicator of leverage was positively correlated, this is not a good phenomenon for first-mover companies. It means that in the process of importing AI-related technology, the forerunners must bear more related costs than second movers. In addition, we can observe that the profit margin showed a positive impact, with the index of 0.137, which means that after the implementation of AI by pioneer companies, the profit improvement was about 13.7%, resulting in a competitive advantage of first movers.

All in all, first movers bore the most relevant cost expenditures in the early stage, and faced the problem of wasting enterprise resources, but they could improve profit and gain a competitive advantage.

Table 8 shows the influence of being a first mover on productivity. The formula (4–2) shows that the value added and output are 0.065,

TABLE 7 | Performance ratio regressions (samples limited to AI adopters only).

Interpretation	Labor productivity	Return on assets (ROA)	Inventory turnover	Return on equity (ROE)	Profit margin	Assets utilization	Collection efficiency	Leverage
Dependent variable	<i>In(Sales)</i>	<i>In(Pretax Income)</i>	<i>In(Cost of Goods Sold)</i>	<i>In(Pretax Income)</i>	<i>In(Pretax Income)</i>	<i>In(Sales)</i>	<i>In(Sales)</i>	<i>In(Debt)</i>
<i>Implementation</i>	−0.31 (0.083)	0.027 (0.054)	0.04 (0.163)	0.093 (0.102)	0.137 (0.116)	−0.061 (0.119)	−0.053 (0.140)	0.048 (0.12)
<i>In(Employees)</i>	0.871*** (0.066)							
<i>In(Assets)</i>		0.082*** (0.035)				0.709*** (0.082)		
<i>In(Inventory)</i>			0.685*** (0.079)					
<i>In(Equity)</i>				0.805*** (0.075)				0.817*** (0.087)
<i>In(Sales)</i>					0.743*** (0.105)			
<i>In(Account Receivable)</i>							0.567*** (0.072)	
Control variable	Industry							
<i>R</i> ²	0.751	0.637	0.453	0.662	0.564	0.464	0.279	0.662
<i>N</i>	50	48	41	46	48	50	50	48

****p* < 0.001.

TABLE 8 | Productivity and market value regressions (samples limited to AI adopters only).

Interpretation	Value added	Output	Tobin's q
Dependent variable	<i>In(Gross Profit)</i>	<i>In(Sale)</i>	<i>In(Market value)</i>
<i>Implementation</i>	0.065 (0.129)	0.08* (0.039)	−0.028 (0.114)
<i>In(Ordinary Capital)</i>	0.211 (0.212)	0.071 (0.064)	
<i>In(Employees)</i>	0.125 (0.255)	0.211* (0.074)	
<i>In(Labor Expense)</i>	0.653 (0.381)	0.556** (0.125)	
<i>In(Cost of Goods Sold)</i>		0.219** (0.032)	
<i>In(Total Assets)</i>			0.741*** (0.078)
Control variable	Industry		
R ²	0.916	0.994	0.54
Observation	14	14	50

* $p < 0.05$.** $p < 0.01$.*** $p < 0.001$.

and 0.08, respectively. The value added increased about 6.5%, and the output increased about 8%. From the results, the impact of being a first mover on productivity was positive, and, in particular, the impact on output was significant and positive. However, fixed costs, such as depreciation and the implementation expenses of new IT systems, tend to offset the associated benefits. The result of the market value shown in Table 8 is −0.028. Although the result shows a negative relationship, the first mover gained advantages due to early entry, and the negative result is not important.

In summary, the Hypothesis 2b results are positively and significant, and thus partially supported, while the Hypotheses 2a and 2c results are not significantly negative, which rejects our assumptions.

5.3 | Comparison Between Better (AI) Performers and Non-Better (AI) Performers

As discussed in the previous section, AI, which is the most popular new IT technology, has been implemented by many industries and improved operational efficiencies, such as decision making and efficient contact with customers. We also found that retail, finance, and insurance are the better performing industries, outperforming the other four industries.

We used formula (3–3) to check whether the better performing industries were better than others according to financial indicators. The results are shown in Table 9. We can see that the results of financial indicators return on assets (−0.152), inventory turnover (−0.076), assets utilization (−1.51), collection efficiency (−0.194), leverage (−0.021) are all negative, indicating that there is no significant impact on financial performance. Hypothesis 3a is thus rejected.

Notably, the Tobin's q (−0.242) result showed a significant negative impact on the market value, as shown in Table 10. This is the opposite of our hypothesis (5–3). In order to understand the

main cause of this conflict, we analyzed the “retail” and “financial and insurance” industries separately. The results of the market value indicators for retail were not significant, so we do not discuss them. Table 11 shows that the financial and insurance industries' Tobin's q (−0.296) is negative and more significant than the previous indicator (−0.242).

As a result, we tried to figure out what caused the market value to be negative and more significant. We found that the years we collected data were from 2010 to 2017, and there was a financial crisis (2008) during that period, which affected people's confidence in the investments of the bank industry. Furthermore, we found a news stories indicating that more than half of the people in the United States still had no confidence in the future of the economy until 2012. They believed that the economy was still in a slump (Morgan 2011). The study of Zouaoui, Nouyrigat, and Beer (2011) proposed that investor sentiments seem to have a positive impact on the likelihood of a stock market crash. For countries culturally prone to collective behavior and overreaction, investor sentiment has a greater impact on the stock market.

Therefore, when companies spend a lot of capital on new IT technologies in an uncertain economy, investors will have a negative and significant response. As a result, Hypotheses 3a and 3c are rejected, yet Hypothesis 3b is partially supported.

6 | Discussion, Implications, Limitations, and Conclusion

In the next section, we discuss our findings, impacts, and research limitations. Through the hypothesis and research framework established earlier, we analyzed the short-term and long-term impact of AI implementation on improvement of performance at the firm-level. We hope that this report can serve as a reference for scholars and managers on the implementation of new IT technologies. Therefore, we provide conclusions and future research recommendations.

TABLE 9 | Performance ratio regressions (samples limited to AI adopters only).

Interpretation	Labor productivity	Return on assets (ROA)	Inventory turnover	Return on equity (ROE)	Profit margin	Assets utilization	Collection efficiency	Leverage
Dependent variable	<i>In(Sales)</i>	<i>In(Pretax Income)</i>	<i>In(Cost of Goods Sold)</i>	<i>In(Pretax Income)</i>	<i>In(Pretax Income)</i>	<i>In(Sales)</i>	<i>In(Sales)</i>	<i>In(Debt)</i>
<i>Implementation</i>	0.01 (0.083)	-0.152 (0.105)	-0.076 (0.165)	0.023 (0.104)	0.037 (0.118)	-1.51 (0.118)	-0.194 (0.142)	-0.21 (0.12)
<i>In(Employees)</i>	0.886*** (0.065)							
<i>In(Assets)</i>		0.847*** (0.074)				0.734*** (0.081)		
<i>In(Inventory)</i>			0.706*** (0.078)					
<i>In(Equity)</i>				0.807*** (0.076)				0.823*** (0.088)
<i>In(Sales)</i>					0.749*** (0.107)			
<i>In(Account Receivable)</i>							0.626*** (0.073)	
Control variable	Industry							
<i>R</i> ²	0.739	0.647	0.485	0.638	0.565	0.504	0.338	0.675
<i>N</i>	49	48	41	46	48	50	50	48

****p* > 0.001.

TABLE 10 | Productivity and market value regressions (samples limited to AI adopters only).

Interpretation	Value added	Output	Tobin's q
Dependent variable	<i>In(Gross Profit)</i>	<i>In(Sale)</i>	<i>In(Market value)</i>
<i>Implementation</i>	0.391* (0.321)	0.128 (0.132)	−0.242** (0.108)
<i>In(Ordinary Capital)</i>	−0.273 (0.253)	−0.22 (0.101)	
<i>In(Employees)</i>	0.465 (0.234)	0.313* (0.096)	
<i>In(Labor Expense)</i>	0.852* (0.312)	0.521** (0.131)	
<i>In(Cost of Goods Sold)</i>		0.281*** (0.034)	
<i>In(Total Assets)</i>			0.798*** (0.074)
Control variable	Industry		
R ²	0.948	0.993	0.594
Observation	14	14	50

p* < 0.05.*p* < 0.01.****p* < 0.001.**TABLE 11** | Productivity and market value regressions (samples limited to AI adopters only).

Interpretation	Value added	Output	Tobin's q
Dependent variable	<i>In(Gross Profit)</i>	<i>In(Sale)</i>	<i>In(Market value)</i>
<i>Implementation</i>	0.391* (0.321)	0.128 (0.132)	−0.296** (0.112)
<i>In(Ordinary Capital)</i>	−0.273 (0.253)	−0.22 (0.101)	
<i>In(Employees)</i>	0.465 (0.234)	0.313** (0.096)	
<i>In(Labor Expense)</i>	0.852* (0.312)	0.521** (0.131)	
<i>In(Cost of Goods Sold)</i>		0.281*** (0.034)	
<i>In(Total Assets)</i>			0.838*** (0.075)
Control variable	Industry		
R ²	0.948	0.993	0.616
Observation	14	14	50

p* < 0.05.*p* < 0.01.****p* < 0.001.

6.1 | Observation

6.1.1 | Inspection of AI Adopters

Through regression testing, the results presented for financial indicators are better for AI adopters. The positive and significant findings show that AI adopters have better market value. As for productivity, adopting AI has positive, but not significant impact, which means that the support is not as strong as with the former two indicators.

6.1.2 | Inspection of First Movers

Through regression testing, the results showed that being a first mover did not have a positive and significant impact on financial

indicators, and almost half of them were negative. As for market value, although we found a negative result, it had little effect because of the advantages of being a first mover. For productivity, the results showed a positive and significant impact.

6.1.3 | Inspection of Better Performers

Through regression testing, the results showed that being a better performer did not have a positive and significant impact on financial indicators, and most of them were negative. In terms of productivity, there was a positive and significant impact on the value added. However, the impact on market value was negative and significant. We showed that the impact came from the financial and insurance industries. In conclusion, we suggest that companies investing heavily in new IT technologies amidst

economic uncertainty may have contributed to the negative and significant reactions from investors.

6.2 | Implications

The latest advancements in AI have enhanced the performance evaluation of firms. AI implementation is diverse and has a revolutionary impact on the firm level. It can not only improve the performance of companies in many ways, but also lead to innovation of business models. Because of the huge changes in new IT implementation, many scholars have also discussed the impact. However, there has been a lack of research on the implementation of AI. Therefore, to provide a comprehensive discussion, we applied a variety of methods to measure the impact of AI implementation on internal performance.

This survey confirms that AI is beneficial and provides a competitive advantage to companies. Our findings offer practical value for managers and enterprises. However, firms in the financial and insurance industries should exercise greater caution when investing in AI.

6.3 | Limitations and Future Research

The first limitation is data analysis. As mentioned above, we collected data from 2010 to 2017, and the base year for the research was 2017. We did not carry out a longitudinal study for several years. Since the benchmarks are the same, this approach does not provide a comprehensive estimate over time. Each company implemented AI at different times, so we should consider each fiscal year. Therefore, we recommend that researchers can use average performance data from different years as a benchmark for investigating AI performance. This would lead to more accurate results.

Second, the firm data structure of this study comes from the S&P 500 index list. The data we used in this study were based on April 30, 2019. However, the S&P 500 Index list is not fixed and can be adjusted at any time. Therefore, while the lists used by other researchers were lightly different from ours, this did not affect our research results. In addition, to gather information on companies' implementation of AI-related initiatives, we sourced data from official websites, academic journals, magazines, and Google, including details on when each company began adopting AI. Some companies implemented AI but did not show relevant information on the Internet, so that we were unable to collect their information. This may have influenced the accuracy of our experimental results. Google search results also revealed that many companies not listed on the S&P 500 have begun investing in AI-related operations; however, these companies were not included in our dataset. Therefore, we recommend that future researchers can extend the sample size from the S&P 500 to the S&P 1000 to ensure the external validity of research results.

6.4 | Conclusion

This study has examined the impact of AI implementation on firm performance through a comprehensive analysis of financial

performance, productivity measurement, and market value (Tobin's q). Focusing on firms outside the IT-centric sector, we have explored the tangible effects of AI adoption, first-mover advantage, and better performance metrics on AI-enabled firms. Our findings reveal that while AI adoption generally enhances financial performance and market value, the anticipated benefits for first movers and better performers are not uniformly realized across all indicators. This insight offers valuable implications for managers and stakeholders, emphasizing a nuanced approach to AI investment decisions that considers both potential gains and limitations.

Looking forward, this study contributes to a deeper understanding of AI business value and opens up several future prospects. First, future research could expand the sample scope to include additional sectors and international firms, providing a broader perspective on AI global impact. Longitudinal studies would also be valuable to track AI effects over time, particularly to observe any evolving productivity impacts and the persistence of market value gains. In addition, as AI technologies rapidly advance, exploring the specific contributions of emerging AI types, such as generative AI or predictive analytics, to different aspects of firm performance would provide further insights. These directions for future research will help managers and stakeholders better understand the long-term business potential of AI and refine strategic approaches to AI integration.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Endnotes

¹ <https://www.sas.com/sas/offers/18/ai-momentum-maturity-success-models-109926.html>.

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